

Sequences & Progressions

Comprehensive All-in-One Concept Guide & Solved Question Bank

Prepared for Class 9th Students • Formatted with LaTeX for Maximum Readability

1 Conceptual Understanding (The "Why")

Patterns exist everywhere around us—in nature, art, music, finance, and everyday life. They help us make sense of our world and predict what comes next. In mathematics, we use sequences, progressions, and series to formalize these patterns.

1.1 1. Defining the Core Terms

Key Definitions

- **Sequence:** An **ordered list of numbers** where each number is called a **term**. Subscripts match position numbers (e.g., t_1 is the first term, t_2 is the second, and t_n is the n^{th} term). Sequences can be **finite** (containing a fixed number of terms, like 6, 12, 24, 48, 96) or **infinite** (continuing indefinitely, written with three dots $\dots$, like 1, 2, 3, 4, 5, 6, $\dots$).
- **Progression:** A special type of sequence whose terms follow a specific, predictable mathematical rule.
- **Series:** The algebraic sum of the terms of a sequence. For example, if a sequence is 1, 3, 5, 7, the corresponding series is $1 + 3 + 5 + 7$.

1.2 2. Types of Rules: Explicit vs. Recursive

To define a sequence precisely, we describe its pattern using a mathematical rule:

- **Explicit Rule (or Explicit Formula):** A rule that uses the term's position number, n , to calculate its value directly. It allows us to find any term (like the 100^{th} or 1000^{th} term) instantly without knowing the previous terms.
 - *Example:* For the odd numbers sequence, the explicit rule is $u_n = 2n - 1$. Testing $n = 3$ gives $u_3 = 2(3) - 1 = 5$.
- **Recursive Rule (or Recursive Formula):** A rule that defines a term by relating it to the previous term(s). To use a recursive rule, the first term(s) must be specified.
 - *Example:* The sequence 1, 4, 7, 10, 13, $\dots$ has a recursive rule of $t_1 = 1$, and $t_n = t_{n-1} + 3$ for $n \geq 2$.

1.3 3. The Historical Virahāṅka-Fibonacci Sequence

A recursive rule can involve multiple previous terms. The most famous example is the **Virahāṅka-Fibonacci sequence**, where each term is the sum of the previous two terms:

The Virahāṅka-Fibonacci Recurrence Relation

$$V_1 = 1, \quad V_2 = 2, \quad V_n = V_{n-1} + V_{n-2} \quad \text{for } n \geq 3$$

This generates the sequence: 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, ...

History of Discovery: While European literature often attributes this to the Italian mathematician Fibonacci (c. 1200 CE), Indian mathematicians discovered and analyzed this pattern centuries earlier:

- **Virahāṅka (7th Century CE):** First explicitly wrote down and studied this sequence in his work *Vṛttajāṭisamuchaya*. He discovered the pattern while analyzing **Prakrit meter and poetry** (relating to long and short syllables).
- **Gopāla (c. 1135 CE):** Further studied the sequence and its properties.
- **Hemachandra (c. 1150 CE):** Linguist-mathematician who thoroughly analyzed and documented the sequence.

2 Arithmetic Progressions (AP)

An **Arithmetic Progression (AP)** is a sequence of numbers in which each term after the first is obtained by adding a fixed number d to the previous term.

2.1 1. Fundamental Parameters of an AP

- **First Term (a):** The starting number of the AP.
- **Common Difference (d):** The constant difference between any consecutive terms (i.e., $d = t_n - t_{n-1}$). The common difference d can be positive, negative, or zero.

2.2 2. Deriving the Explicit Formula Logically

Let's build an AP term-by-term to see the pattern:

- **1st Term (t_1):** a
- **2nd Term (t_2):** $a + d = a + 1 \times d$
- **3rd Term (t_3):** $(a + d) + d = a + 2d = a + 2 \times d$
- **4th Term (t_4):** $(a + 2d) + d = a + 3d = a + 3 \times d$

Notice that for any term at position n , the number of times we add d is exactly **one less than its position number** ($n - 1$).

Arithmetic Progression (AP) Formulas

Explicit Formula for the n^{th} term:

$$t_n = a + (n - 1)d$$

Recursive Formula for the n^{th} term:

$$t_1 = a, \quad t_n = t_{n-1} + d \quad \text{for } n \geq 2$$

2.3 3. Visualizing an AP

When we form ordered pairs (x, y) where x represents the term position n and y represents the term value t_n , and plot them on a coordinate grid, **the points always lie on a straight line**. Because the rate of change is constant (d), an AP represents a **linear pattern**.

2.4 4. Sum of the First n Natural Numbers (S_n)

The sum of the first n natural numbers is $S_n = 1 + 2 + 3 + \cdots + n$. We can derive an elegant formula using the "reversal" method:

$$S_n = 1 + 2 + \cdots + (n - 1) + n$$

$$S_n = n + (n - 1) + \cdots + 2 + 1$$

Adding these two equations column-by-column:

$$2S_n = (n + 1) + (n + 1) + \cdots + (n + 1) \quad [n \text{ times}]$$

$$2S_n = n(n+1) \implies S_n = \frac{n(n+1)}{2}$$

Historical Context: The first known written mention of this formula is found in **Āryabhaṭa's Āryabhaṭīya** (Chapter 2, Verse 19, written in 499 CE). He described that the sum is calculated by taking the average of the first and last terms and multiplying by the total number of terms: $S_n = n \times \frac{a+l}{2}$.

3 Geometric Progressions (GP)

A **Geometric Progression (GP)** is a sequence of numbers in which each term after the first is obtained by multiplying the previous term by a fixed, non-zero number r .

3.1 1. Fundamental Parameters of a GP

- **First Term (a):** The starting number of the GP.
- **Common Ratio (r):** The constant multiplier found by dividing any term by its preceding term (i.e., $r = \frac{t_n}{t_{n-1}}$).

3.2 2. Deriving the Explicit Formula

Let's build a GP term-by-term:

- **1st Term (t_1):** a
- **2nd Term (t_2):** $a \times r = ar^1$
- **3rd Term (t_3):** $(ar) \times r = ar^2$
- **4th Term (t_4):** $(ar^2) \times r = ar^3$

Observe that the exponent of r is always **one less than the term number** ($n - 1$).

Geometric Progression (GP) Formulas

Explicit Formula for the n^{th} term:

$$t_n = ar^{n-1}$$

Recursive Formula for the n^{th} term:

$$t_1 = a, \quad t_n = r \times t_{n-1} \quad \text{for } n \geq 2$$

3.3 3. Visualizing a GP

If we plot the ordered pairs (n, t_n) for a GP, **the points do not lie on a straight line**. Instead, they form a curve that grows or decays rapidly (exponential curve).

3.4 4. Real-World Applications & Case Studies

3.4.1 A. Fractals: The Sierpiński Triangle & Square Carpet

Many properties of fractals (patterns that repeat at different scales) lead directly to GPs:

- **Sierpiński Triangle:** Created by repeatedly removing the middle equilateral triangle from Stage 0 (area = 1).
 - **Number of Shaded Triangles:** Forms a GP: 1, 3, 9, 27, ... given by $t_n = 3^n$ (Stage n).
 - **Area of Shaded Region:** Forms a GP with common ratio $r = \frac{3}{4}$: $s_n = \left(\frac{3}{4}\right)^n$. As n increases, the shaded area rapidly diminishes and approaches 0.

- **Sierpiński Square Carpet:** A square sheet of paper is divided into 9 smaller squares, and the center is removed. The remaining 8 squares are retained. The process repeats.
 - **Number of Red/Shaded Squares:** Forms a GP: $1, 8, 64, 512, \dots$ given by $t_n = 8^n$.
 - **Area of Shaded Region:** Forms a GP with ratio $r = \frac{8}{9}$: $s_n = \left(\frac{8}{9}\right)^n$.

3.4.2 B. Physics: Bouncing Ball Heights

When a ball is dropped from an initial height, its bounce heights form a decaying GP. For example, if dropped from 24 feet and bouncing back to $\frac{3}{4}$ of its previous height, the bounce heights form a GP with $a = 18$ (first bounce height) and $r = 0.75$.

4 Essential Formulas Quick-Reference Sheet

Use this structured reference table to quickly select the correct formula for any problem.

Formula Type	Mathematical Formula	Variables & Parameter Definitions
AP n^{th} Term (Explicit)	$t_n = a + (n - 1)d$	a : first term, d : common difference, n : term position
AP Recursive Rule	$t_n = t_{n-1} + d$	t_{n-1} : preceding term, d : common difference
GP n^{th} Term (Explicit)	$t_n = ar^{n-1}$	a : first term, r : common ratio, n : term position
GP Recursive Rule	$t_n = r \times t_{n-1}$	t_{n-1} : preceding term, r : common ratio
Sum of First n Naturals	$S_n = \frac{n(n+1)}{2}$	S_n : total sum, n : count of consecutive natural numbers
AP Sum (Average Rule)	$S_n = n \times \frac{a+l}{2}$	a : first term, l : last term, n : number of terms

Table 1: Core Formulations for Sequences and Progressions

5 Problem-Solving Strategy: Worked Question Bank

This section contains detailed, step-by-step solutions to twelve fundamental problem types. Each problem is designed to demonstrate key mathematical strategies.

5.1 1. Arithmetic Progression (AP) Problems

Question 1: Finding Specific Terms

Question: Find the 10th and 26th terms of the AP: 3, 8, 13, 18, ...

Key Concept: Identifying parameters and applying the explicit term formula: $t_n = a + (n - 1)d$.

Step-by-Step Solution:

1. **Identify a and d :** The first term is $a = 3$. The common difference is $d = 8 - 3 = 5$.

2. **Find t_{10} ($n = 10$):**

$$t_{10} = 3 + (10 - 1) \times 5 = 3 + 9(5) = 3 + 45 = 48$$

3. **Find t_{26} ($n = 26$):**

$$t_{26} = 3 + (26 - 1) \times 5 = 3 + 25(5) = 3 + 125 = 128$$

Answer: The 10th term is **48** and the 26th term is **128**.

Question 2: Solving for Term Position

Question: Which term of the AP: 21, 18, 15, ... is -81 ? Also, is 0 a term of this AP?

Key Concept: Setting t_n to a target value and checking if the resulting n is a natural number ($n \in \mathbb{N}$).

Step-by-Step Solution:

1. **Identify a and d :** The first term is $a = 21$. The common difference is $d = 18 - 21 = -3$.

2. **Solve for -81 :** Set $t_n = -81$ and solve:

$$-81 = 21 + (n - 1)(-3)$$

$$-81 - 21 = -3(n - 1) \implies -102 = -3(n - 1)$$

$$n - 1 = \frac{-102}{-3} = 34 \implies n = 35$$

3. **Solve for 0:** Set $t_n = 0$:

$$0 = 21 + (n - 1)(-3) \implies -21 = -3(n - 1)$$

$$n - 1 = \frac{-21}{-3} = 7 \implies n = 8$$

Answer: -81 is the 35th term, and 0 is the 8th term of the AP.

Question 3: Formulating General and Recursive Rules

Question: Find the n^{th} term and the recursive rule for the AP: 11, 8, 5, 2, ...

Key Concept: Translating specific parameters into general formulas.

Step-by-Step Solution:

1. **Identify a and d :** The first term is $a = 11$. The common difference is $d = 8 - 11 = -3$.

2. **Write the Explicit Formula:**

$$t_n = 11 + (n - 1)(-3) = 11 - 3n + 3 = 14 - 3n$$

3. **Write the Recursive Rule:** State the first term and the relation:

$$t_1 = 11, \quad t_n = t_{n-1} - 3 \quad \text{for } n \geq 2$$

Answer: The explicit formula is $t_n = 14 - 3n$ and the recursive rule is $t_1 = 11, t_n = t_{n-1} - 3$.

Question 4: Solving Simultaneous AP Systems

Question: Find the 31st term of an AP whose 11th term is 38 and 16th term is 73.

Step-by-Step Solution:

1. **Set up the system of linear equations:**

$$\text{Eq 1: } t_{11} = a + 10d = 38$$

$$\text{Eq 2: } t_{16} = a + 15d = 73$$

2. **Subtract Eq 1 from Eq 2:**

$$(a + 15d) - (a + 10d) = 73 - 38 \implies 5d = 35 \implies d = 7$$

3. **Substitute $d = 7$ into Eq 1:**

$$a + 10(7) = 38 \implies a + 70 = 38 \implies a = -32$$

4. **Calculate t_{31} :**

$$t_{31} = a + 30d = -32 + 30(7) = -32 + 210 = 178$$

Answer: The 31st term is **178**.

Question 5: Excess Conditions

Question: Determine the AP whose third term is 16 and whose 7th term exceeds the 5th term by 12.

Key Concept: Interpreting relative algebraic relations to isolate the common difference.

Step-by-Step Solution:

1. **Formulate the excess equation:**

$$t_7 = t_5 + 12$$

$$a + 6d = (a + 4d) + 12$$

2. **Solve for d :** Subtract a and $4d$ from both sides:

$$2d = 12 \implies d = 6$$

3. **Solve for a using the third term:**

$$t_3 = a + 2d = 16 \implies a + 2(6) = 16$$

$$a + 12 = 16 \implies a = 4$$

4. **Construct the AP:** Start with $a = 4$ and add $d = 6$ repeatedly: 4, 10, 16, 22, ...

Answer: The required AP is 4, 10, 16, 22, ...

Question 6: Divisibility in Three-Digit Numbers

Question: How many three-digit numbers are divisible by 7?

Key Concept: Establishing the lower and upper boundaries of a finite AP within a specific range.

Step-by-Step Solution:

1. **Find the first term (a):** The smallest three-digit number is 100. $100 \div 7 \approx 14.28$. The first multiple of 7 in this range is $7 \times 15 = 105$. So, $a = 105$.
2. **Find the last term (t_n):** The largest three-digit number is 999. $999 \div 7 \approx 142.71$. The largest multiple of 7 is $7 \times 142 = 994$. So, $t_n = 994$.
3. **Use the AP explicit formula to solve for n ($d = 7$):**

$$994 = 105 + (n - 1) \times 7$$

$$994 - 105 = 7(n - 1) \implies 889 = 7(n - 1)$$

$$n - 1 = \frac{889}{7} = 127 \implies n = 128$$

Answer: There are exactly **128** three-digit numbers divisible by 7.

Question 7: Multiples of 4 within a Bound

Question: How many multiples of 4 lie between 10 and 250? (Note: "Between" means we exclude 10 and 250 themselves from being boundaries unless they fall strictly inside the interval).

Step-by-Step Solution:

1. **Identify first term (a):** The first multiple of 4 strictly greater than 10 is $a = 12$.
2. **Identify last term (t_n):** The largest multiple of 4 strictly less than 250 is $t_n = 248$. (Note: We cannot use 256 or 252 because they fall outside the 250 ceiling).
3. **Use the explicit formula with $d = 4$ to solve for n :**

$$248 = 12 + (n - 1) \times 4 \implies 236 = 4(n - 1)$$

$$n - 1 = \frac{236}{4} = 59 \implies n = 60$$

Answer: There are **60** multiples of 4 that lie between 10 and 250.

Question 8: Divisibility by 9 Challenge

Question: How many three-digit numbers are divisible by 9? Why is dividing 999 by 9 not the direct answer?

Key Concept: Recognizing that $999 \div 9 = 111$ includes single-digit and double-digit multiples of 9 starting from 1. We must filter out the non-three-digit numbers.

Step-by-Step Solution:

1. **Find first term (a):** Smallest three-digit multiple of 9 is $a = 108$ (9×12).
2. **Find last term (t_n):** Largest three-digit multiple of 9 is $t_n = 999$ (9×111).
3. **Solve for n with $d = 9$:**

$$999 = 108 + (n - 1) \times 9 \implies 891 = 9(n - 1)$$

$$n - 1 = \frac{891}{9} = 99 \implies n = 100$$

Alternative Verification Method: There are 111 multiples of 9 from 1 to 999. The multiples below 100 (one-digit and two-digit numbers) range from $9 \times 1 = 9$ to $9 \times 11 = 99$ (exactly 11 terms). Excluding them gives: $111 - 11 = 100$. Both methods yield identical results, but the AP method is more direct.

Answer: There are **100** three-digit numbers divisible by 9.

Question 9: Double-Digit Divisibility and Summation

Question: How many 2-digit numbers are divisible by 3? What is the sum of all these 2-digit numbers?

Key Concept: Combining the count of terms (n) with the summation formula S_n .

Step-by-Step Solution:

1. **Identify first term (a), last term (t_n), and difference (d):** The smallest 2-digit multiple of 3 is $a = 12$. The largest is $t_n = 99$. The common difference is $d = 3$.

2. **Find the count of terms (n):**

$$99 = 12 + (n - 1) \times 3 \implies 87 = 3(n - 1)$$

$$n - 1 = 29 \implies n = 30$$

3. **Calculate the sum (S_{30}):** Use Āryabhaṭa's average-based summation formula:

$$S_{30} = n \times \left(\frac{a + t_n}{2} \right) = 30 \times \left(\frac{12 + 99}{2} \right) = 15 \times 111 = 1665$$

Answer: There are **30** two-digit numbers divisible by 3, and their sum is **1,665**.

Question 10: Summation Inequality Challenge

Question: Find the smallest value of n such that the sum of the first n natural numbers is greater than 1,000.

Step-by-Step Solution:

1. **Write the inequality using the natural numbers sum formula:**

$$S_n = \frac{n(n+1)}{2} > 1000 \implies n(n+1) > 2000$$

2. **Estimate n by taking the square root of 2000:**

$$\sqrt{2000} \approx 44.72$$

3. **Test integers near 44.72:**

- Let $n = 44$: $44 \times 45 = 1980$ (This is ≤ 2000).
- Let $n = 45$: $45 \times 46 = 2070$ (This is > 2000).

4. **Verify the sum for $n = 45$:**

$$S_{45} = \frac{45 \times 46}{2} = 1,035 > 1000$$

Answer: The smallest value of n is **45**.

5.2 2. Geometric Progression (GP) Problems

Question 11: Solving GP Sum and Product Systems

Question: The sum of the first three terms of a GP is $\frac{13}{12}$ and their product is -1 . Find the common ratio and the terms.

Key Concept: Representing the three terms as $\frac{a}{r}, a, ar$ to simplify product computations.

Step-by-Step Solution:

1. **Formulate the product equation:**

$$\left(\frac{a}{r}\right) \times a \times (ar) = -1 \implies a^3 = -1 \implies a = -1$$

2. **Formulate the sum equation with $a = -1$:**

$$\frac{-1}{r} - 1 - r = \frac{13}{12} \implies \frac{-1}{r} - r = \frac{13}{12} + 1 = \frac{25}{12}$$

Multiply the entire equation by $-12r$ to eliminate fractions and rearrange:

$$12 + 12r^2 = -25r \implies 12r^2 + 25r + 12 = 0$$

3. **Solve the quadratic equation by factoring:**

$$12r^2 + 16r + 9r + 12 = 0 \implies 4r(3r + 4) + 3(3r + 4) = 0$$

$$(4r + 3)(3r + 4) = 0 \implies r = -\frac{3}{4} \quad \text{or} \quad r = -\frac{4}{3}$$

4. **Calculate the terms for both ratios:**

- **Case 1** ($r = -3/4$): Terms are $\frac{-1}{-3/4}, -1, -1(-3/4) \implies \frac{4}{3}, -1, \frac{3}{4}$
- **Case 2** ($r = -4/3$): Terms are $\frac{-1}{-4/3}, -1, -1(-4/3) \implies \frac{3}{4}, -1, \frac{4}{3}$

Answer: The common ratio is $-\frac{3}{4}$ or $-\frac{4}{3}$, and the terms are $\frac{4}{3}, -1, \frac{3}{4}$.

Question 12: Finding Term Position in a GP

Question: Which term of the GP: 2, 8, 32, ... is 131072? Write the explicit and recursive rules.

Key Concept: Equating the GP explicit formula to a target value and using logarithms/prime factorization.

Step-by-Step Solution:

1. **Identify a and r :** The first term is $a = 2$. The common ratio is $r = 8 \div 2 = 4$.

2. **Set up the explicit equation:**

$$t_n = ar^{n-1} \implies 131072 = 2 \times 4^{n-1}$$

3. **Divide by 2 and express as powers of 2:**

$$65536 = 4^{n-1}$$

Since $4 = 2^2$ and $65536 = 2^{16}$:

$$2^{16} = (2^2)^{n-1} = 2^{2n-2}$$

4. **Equate exponents and solve for n :**

$$16 = 2n - 2 \implies 2n = 18 \implies n = 9$$

5. **Write the general rules:**

- **Explicit Rule:** $t_n = 2 \times 4^{n-1}$
- **Recursive Rule:** $t_1 = 2, \quad t_n = 4t_{n-1} \quad \text{for } n \geq 2$

Answer: 131072 is the 9th term of the GP. The explicit formula is $t_n = 2 \times 4^{n-1}$ and the recursive rule is $t_1 = 2, t_n = 4t_{n-1}$.